Strategy Learner - Classification Learner Hints and Tips

Classification Trader Hints and Tips

Overview

You will refactor your ensemble learner (i.e., regression Bag Learner comprised of DTLearners) to serve as a Random Forest learner (i.e., a classification Bag Learner comprised of RTLearners). You will use your Random Forest learner to train and test a learning trading algorithm. Here are some ideas (gathered from a previous project) that you might find helpful if you are going to use a classification or regression learner for your trader.

ML Trader

One of the first decisions you will need to make is what a Y Target, or prediction, represents. Some common alternatives include using Y to represent trading *actions*, such as `[+1 = Buy, 0 = Do Nothing, -1 = Sell]` or using Y to represent *positions*, such as `[+1 = Long Position, 0 = Cash Position, -1 = Short Position]`. You may need to experiment as you might find that one representation works better in your architecture than the other.

Conceptually, one can think of using the indicators to make a prediction of what the market will look like N days into the future. Based on 1) the indicators on day D, 2) the price on day D, and 3) the price on day D + N, you will use this information to determine the value of Y. For example, if the price on today (i.e., day D) is $100, and the price five days (i.e., where N=5) is $101, could determine that our position should be long. In other words, your derived Y value is +1, which represents going long. We do this for every example in the dataset to build the Y values.

Y Target Algorithm

Your code should classify based on N day change in price. You need to build a new Y that reflects the N day change and aligns with the current date. The algorithm for determining the Y Target (or prediction) values is provided below.

```
# This pseudo code demonstrates the algorithm for determining the Y prediction on a given date.
# It considers looks at the current price (on the current day) and the price N days into the future.
# If this future price is higher, then you might want the learner to be in a long position to take advantage of the rising price
# If this future price is lower, then you might want the learner to be in a short position to take advantage of the falling price

set ret to (price[t+N]/price[t]) - 1.0     #Better to use the N day return than an absolute difference in price.

if ret is greater than YBUY:
        set Y[t] to +1  # LONG
else if ret is less than YSELL:
        set Y[t] to -1  # SHORT
else:
        set Y[t] to  0  # CASH
```

You may want to modify this algorithm if you want the trigger to be above or below a certain amount, which is controlled by changing the values of YBUY and YSELL.

Example of Building the Dataset

Notice how each row in the dataset consists of the 3 to 5 indicator values for a given date and the derived Y value (which could represent a position or an action). Indicator values could be their raw values, their transformed values, or their categorical values. Once again, this may be something where some experimentation is required to determine which works best with your architecture.

		Features (X)	Target (y)

Note that your X indicator values are calculated each day from the current day's (and earlier) data, but the Y value (classification) is calculated using data from the return.

Random Forest Hyperparameters

You may tweak various parameters of your learner to maximize return. Train and test your learning strategy over the in sample period.

The Random Forest learner is controlled using two hyperparameters: leaf_size and number_of_bags. You may need to conduct several research to determine the optimal values of each. You should tweak the parameters of your learner to maximize performance during the in sample period.

- The leaf_size must be equal to or greater than 5 to avoid overfitting
- The number_of_bags must be equal to or greater than 20

Historically, students have performed a grid search for the leaf_size and number_of_bags to find their best values. We do not expect one to use the optimizer to find learner hyperparameters.

Hints & Tips

- For your X values: Identify and implement at least 3 technical features that you believe may be predictive of future return.
- For your Y values: Use future N day return (not future price). Then classify that return as LONG, SHORT or CASH. You're trying to predict a relative change that you can use to invest with.
- For debugging purposes, you may find it helpful to plot the value of the training classification data (-1, 0, 1) versus the stock price in one color.
- For debugging purposes, you may find it helpful to plot the value of the training classification output (-1, 0, 1) versus the stock price in another color. Ideally, these two lines should be very similar.